Welcome to Part One: Notes to Data

In Part One of this tutorial we will learn to:

1. Search Open Context (OC) for a specific data source and download multiple PDF files
2. Open one of the context sheets and evaluate how the observations in it might be transposed
3. Create a table based on the observations in the context sheet
4. Fill in the table based on the observations in the context sheet
5. Select a different context sheet and add those observations to the table

Key concepts used in this tutorial

- data structure
- paperwork or documentation
- context sheets
- columns, attributes, headings
- entries, rows, observations
- data types

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Introduction

Welcome (back) to our digital data stories! In this tutorial, we're going to look at something a bit more unstructured than our usual tutorials. Rather than utilizing a ready-to-go data set, we're going to build one ourselves. Building data sets can be a complicated task. But, we'll walk through all the steps together before trying it on our own.

We may be wondering, why would we do this? With so many archaeological data sets out there, why bother creating our own? Well, as much as we like the well-curated collection of data sets that OC shares, data sets like that are… not that common.

What's more likely is that whenever we start a project, we'll need to decide everything about that project. We'll need to decide how we collect our data, how we observe things, as well as other logistics. These will typically be driven by our research questions but may also be driven by other needs, like when excavation is necessary to preserve something from the site that might be otherwise completely destroyed.

We don't want to mimic a whole excavation through tutorials (at least not yet) so we're skipping some steps to focus on an undervalued, but essential, part of archaeology: paperwork.

After all, we're all here because we love sorting through mounds…of dirt stained pages of notes.

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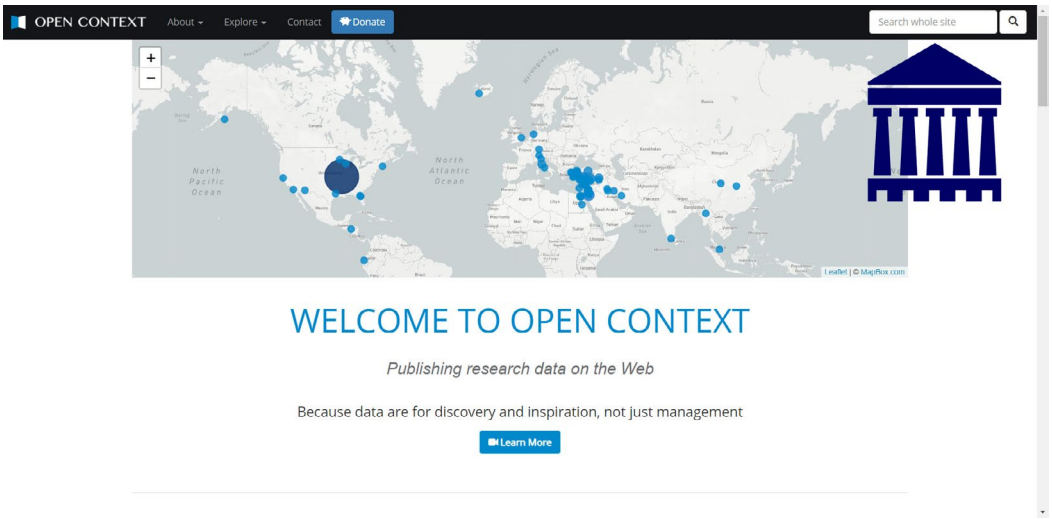
Picking a project

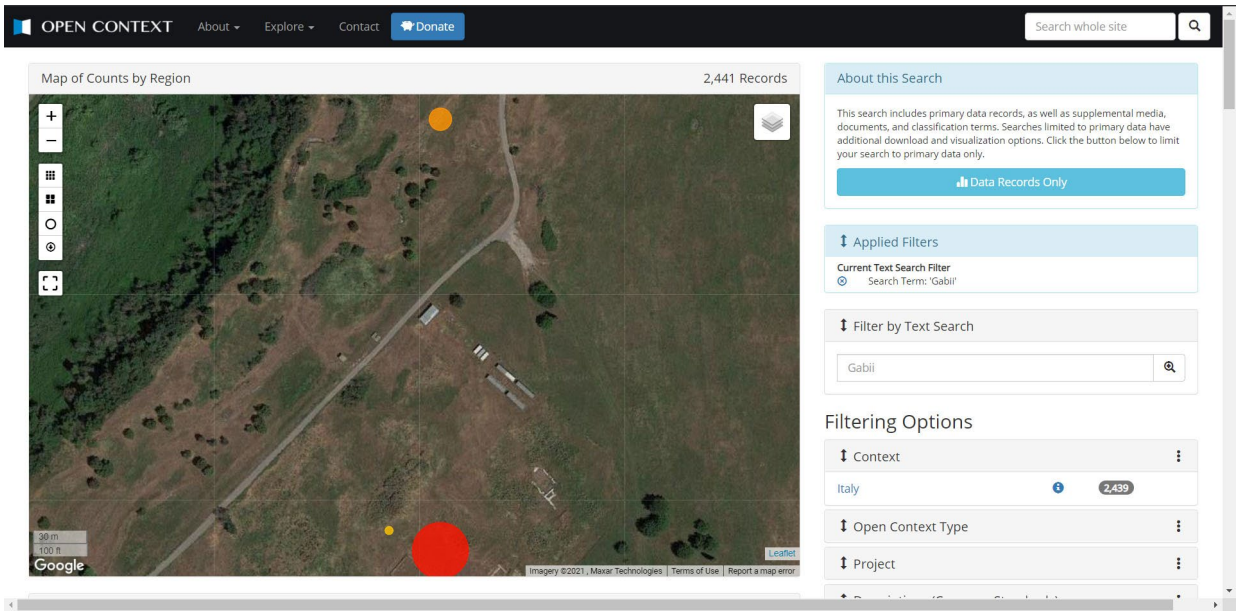
Leaving aside decisions about data collection, we're going to jump to the act of data collation. With that, we can go from Notes (aka paperwork) to Data (aka something to be analyzed) and then from Analyzed Data to Narrative (the story of a site). To do this we'll be using context sheets, a particular form of structured note-taking, from the site of Gabii, which is located in Italy.

But before we do this, we'll need to Rome around OC for this particular project. We'll do this by going to the website first (www.opencontext.org) and putting Gabii into the search bar at the top of the page.



This is where we can find the search bar on the Open Context website.

Once we put Gabii into the search bar, and click search, we should be navigated to a page that looks a bit like this.



Here's what it should look like once we've searched for Gabii. The image may be more or less zoomed out depending on screen resolution and window size, and that's ok!

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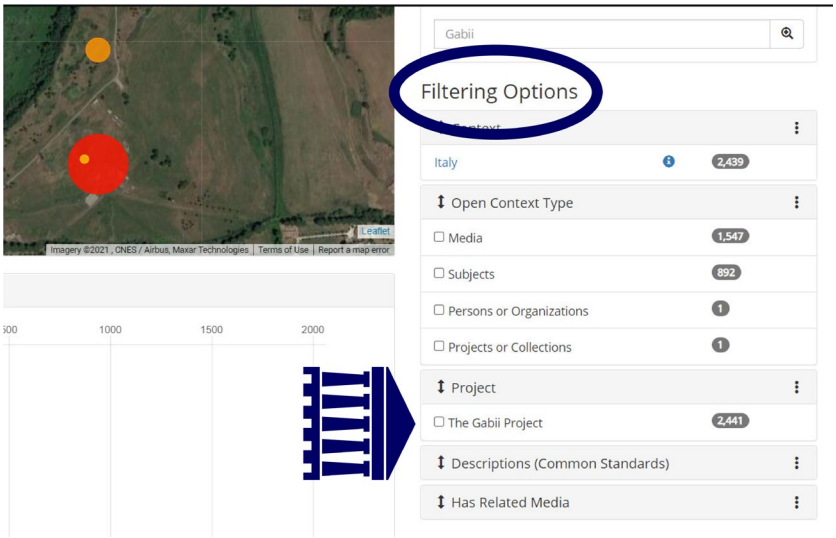
Here, we scroll down to a section called "Filtering Options" and select from several choices.



This is what the filtering options look like.

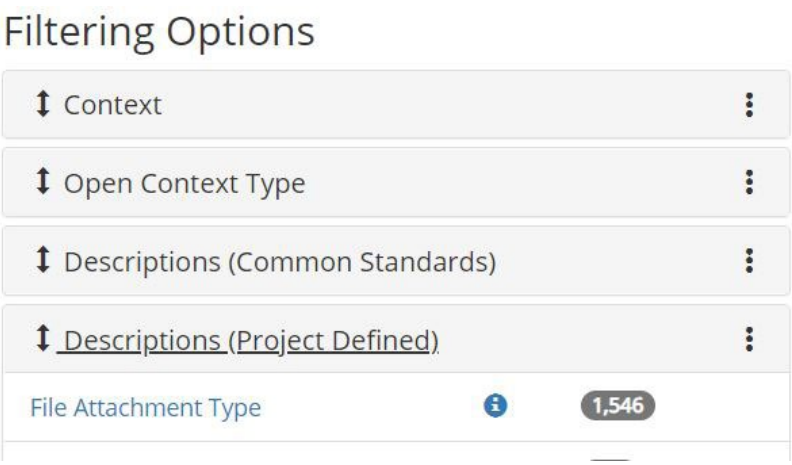
Specifically, we'll want to open the "Project" option and select or click The Gabii Project.

This will take us to another page just for The Gabii Project, which was what we were looking for.

On that page:

1. Scroll down to the filtering options again.
2. Open the "Descriptions (Project Defined)" option.
3. Select or click "File Attachment Type" to filter the data.

This will, once again, bring us to a new page. There, repeat those previous steps but look for the File Attachment Type option.



The filtering option will look like this when we click or select "File Attachment Type".

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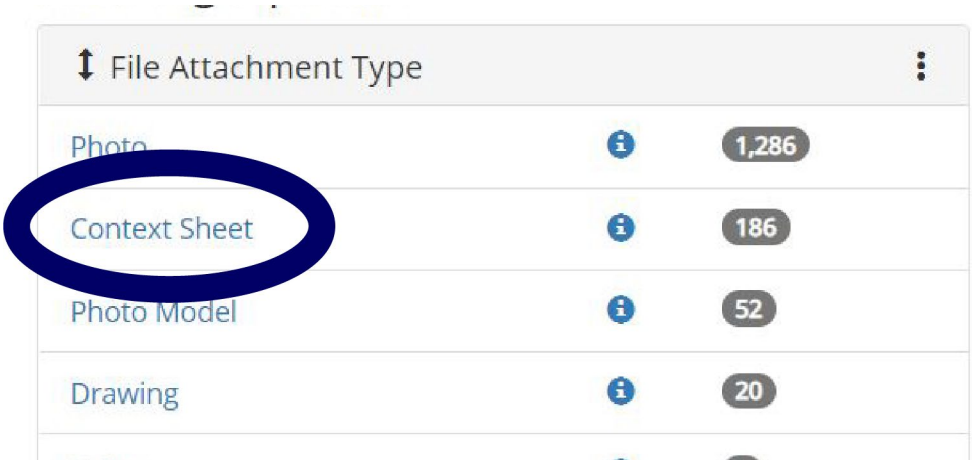
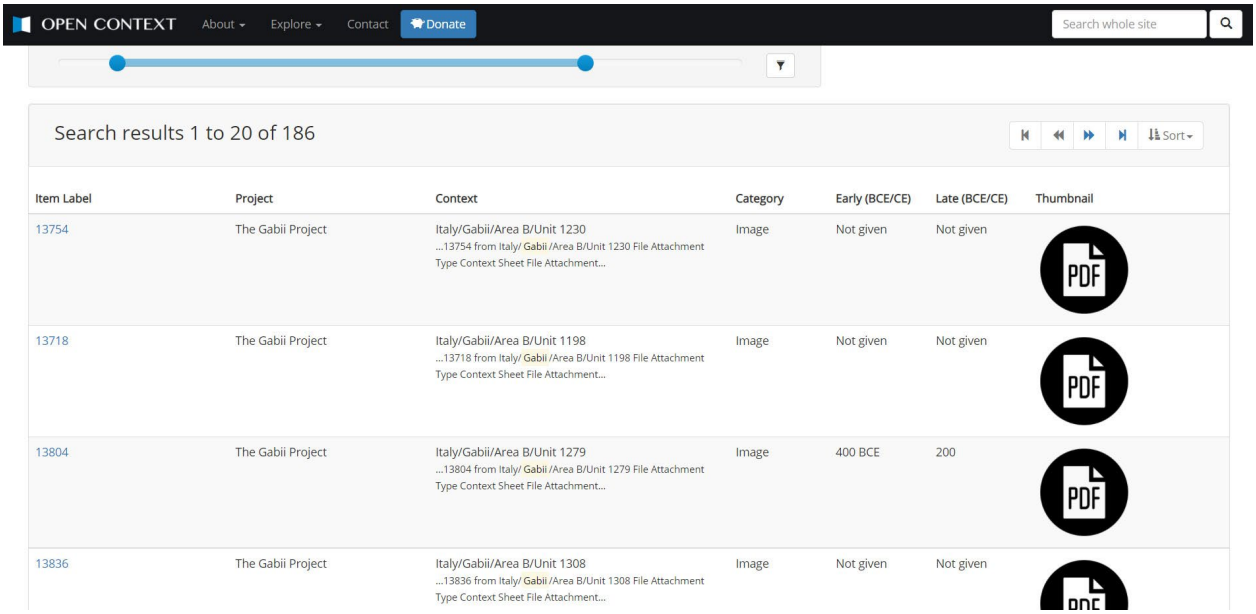
While it might seem tedious to click and scroll, click and scroll, these different layers introduce the many different ways that data in OC are associated, which is part of the power of the tool.



The filtering option for "Context Sheet" will look like this before we click or select it.

Under the "File Attachment Type" filtering option, we'll select the specific kind of file or notes that we want to look at: the Context Sheet. We'll click or select this to get to our final selection.

This last choice will take us to a page with a list of context sheets for The Gabii Project. This time, when we scroll down the page the only data will be the context sheets for the project.



This is what the list of context sheets will look like.

Also, if we want to skip ahead or accidentally mess up the paths or go down an OC rabbit hole, we can go straight to the context sheets from here.

If you need the full URL for ease of use, the context sheets can be found at:

https://opencontext.org/search/?proj=104-the-gabii-project&prop=104-file-attachment-type---104-context-sheet&prop=oc-gen-image&q=Gabii&rows=20&type=media#18/41.88878/12.71939/20/any/Google-Satellite.

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Now that we're here, we want to click on one of the hyperlinked "Item Label" numbers. We suggest selecting the one numbered "13675". If we don't see it on that list immediately, we will need to click the set of ">>" arrows to look at more context sheets. Or we can click the Sort button and select "Item (type, provenance, label), ascending". This will bring number "13675" to the first set of sheets.

Clicking the "13675" link, will take us to that context sheet's unique page. That page will have a button that says "Download File". We'll click that button and it will take us to a view of the PDF. This is where we can download the context sheet to a safe location. Make sure to note where it downloads and under what name we saved it.

This is what the download button will look like on the new page.

Once that's all done, we will have downloaded our first context sheet that we can use to build a data set. For reference, the images for this tutorial are from 13675. However, we can pick any context sheet in the list to start with and will be looking at multiple sheets. So have fun picking different PDFs to explore!

Kinds of data

Awesome! We've downloaded a context sheet. This is an example of one way that The Gabii Project decided to structure their note-taking, by creating a form. It looks like they also took photos, scanned drawings, and collected other data based on the options available under "File Attachment Type" that we saw while we were filtering.

Most projects will choose how to structure their note-taking, and therefore their forms, photos, and drawings, before the project starts. The best projects, though, also think about how those forms will translate into another common structured and organized format, the table, which we may be familiar with if this isn't the first data story we've completed together.

While The Gabii Project already has these data entered in a table or database, the process of turning notes into data still happens regularly in archaeology. This is because many older archaeological projects, sometimes called legacy projects, have not converted their notes to tables. Furthermore, this is also common for projects that don't have access to digital infrastructure in the field.

So, the practice of understanding how to turn a form into a structured table is always important, particularly because context sheets like these include a lot of important observations and data in a short two-page form. Some of these notes will be very easy to turn into a table and some will be harder. However, many of the values that are more difficult to organize can be some of the most important parts of archaeological note-taking.

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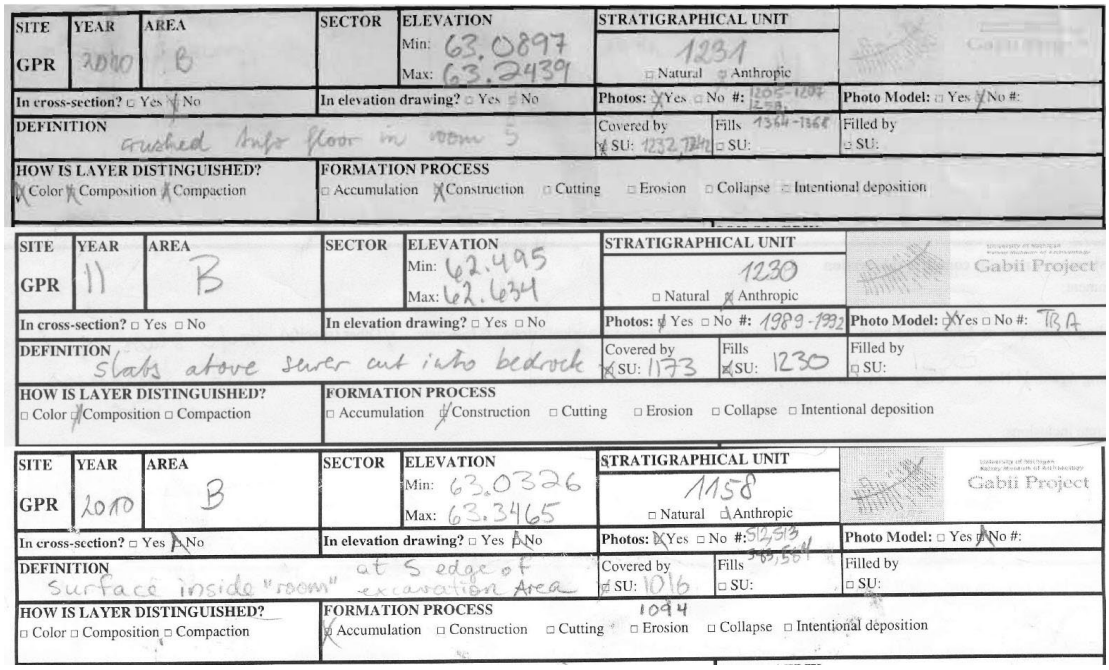
The structure of the context sheet

When we open up our PDF, we'll see that it looks like a…form. It has pencil and pen marks denoting observations about the context (how things are related) and of belongings (also known as artifacts) at the site.

Most archaeological forms have similar styles and what's cool about forms is that rather than just recording whatever the excavator happens to notice, forms guide our observations and formalize what observations we want to make sure we take note of. So take some time to examine the PDF of the context sheet that we downloaded.

- How many different sections are there in this particular form?
- Are they related to each other in any particular way?
- Are some portions not filled out?
- Are some portions crossed out?
- How many of these sections provide options, like lists, checkboxes, or yes/no boxes?
- How many look like they are numeric, in that people only put numbers in the box?
- How many look like someone just wrote whatever?
- Are there any drawings or referenced photographss?
-

When we look at a form, it's important to understand the limits of what it contains. No matter what questions we have, we can't ask them if the excavators didn't note that information. So it's really important to get to know what is there before we move forward. Beyond that, it's important to look at a couple of different sheets before committing to a particular structure for our table.



Here are snippets of three context sheets for comparison. Note the similarities and differences in what is filled out.

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This is because how individuals fill out the forms can influence how we structure our tables and make sure that all the important elements are captured. Furthermore, there are regional differences in handwriting that are important to know. For example, people from European countries might write the number "1" in a way that looks more like how folks in North America write the number "7", while a European "7" will look like a "1", but have a dash mark through it.

However, as we look at these two pages of the 13675 context sheet, there is a lot going on in this form.

…and this is only a tutorial.

We're creating this data set for practice right now, so we don't need to copy every heading because we want to practice organizing the data, not just repeat data entry that's already been done. So instead, we'll focus on a few headings and use these to learn how to turn these notes into structured data!

Parts of a table

There are typically two axes of information that we use when we turn forms into data structured in a table. We can think of each space on the form as a heading, field, column, or attribute for the tabular data we are going to build. These are the specific observations we want to take note of.

The information within that box is the specific entry, row, or observation for the context sheet we're working with. These refer to the specific thing we are observing and the particular instance of the attributes it had. These are what we will enter into our table from each context sheet to start building our data set.

Selecting our headings

After we have a feeling for how the form is organized, we'll need to select our headings. We can keep a list of these headings anywhere, but for the sake of organization it might be a good time to open up our trusted (or not trusted, as all corporations are evil) spreadsheet program to type these up.

Or, get some lined or graph paper and make this into a physical table. Some of our other evaluations might be more difficult to do by hand, for example rewriting descriptions, but transposing data into a physical table can be a great exercise in practicing the mechanics of these transpositions.

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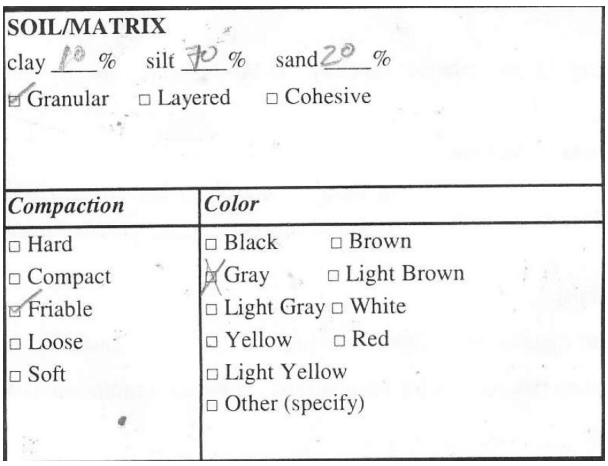
As we let our eyes Rome the form, it looks like the major headings are the **WORDS IN ALL CAPS AND BOLD**, associated with particular boxes on the form. However, some boxes also have ***Words In Camel Caps That Are Italicized And Bold***, as well. These are probably subheadings and we'll want those to be in their own columns. We may also notice that some boxes have some other things going on, like check-boxes.



SOIL/MATRIX	
clay 10 % silt 70 % sand 20 %	
☑ Granular ☐ Layered ☐ Cohesive	
Compaction	***Color***
☐ Hard	☐ Black ☐ Brown
☐ Compact	☒ Gray ☐ Light Brown
☑ Friable	☐ Light Gray ☐ White
☐ Loose	☐ Yellow ☐ Red
☐ Soft	☐ Light Yellow
	☐ Other (specify)

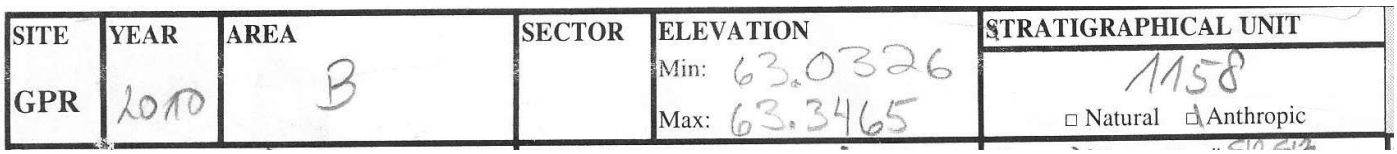
Here's what the SOIL/MATRIX box looks like.

If we take a look at the **SOIL/MATRIX** box, which may or may not be filled in on our context sheet, the section has more than one heading we'll create from it. That's ok!

What we'll want to do, as we're listing out all the headings, is know that there will be multiple headings that will start with soil/matrix. Then they'll have something that denotes specifically what information we're entering. We'll want to make sure we enter these subheadings using a standard pattern as well, so we can quickly see which headings are related.

Good practice in archaeology is to use more column names or headings rather than less. Why? Well, we can always recombine data if we need to. Going in the opposite direction, separating data into multiple columns, is possible BUT typically causes more errors.

This is because the more we type, the more likely we are to cause errors due to spelling or differences in the number of spaces or capitalization.



SITE	YEAR	AREA	SECTOR	ELEVATION	STRATIGRAPHICAL UNIT
GPR	2010	B		Min: 63.0326 Max: 63.3465	1158 ☐ Natural ☑ Anthropic

These are the headings at the top of the context sheet that we'll be using.

Now that we've looked at the various possible boxes we can use as headings, we're going to focus on the ones at the top of the context sheet, and then the **DESCRIPTION** section on the first page, which should have at least two subheadings. Once we've got our list of headings, with more headings probably than the number of boxes we had, we can put them into a table.

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Making our table

Once we have a new spreadsheet document open, or we've located some appropriate paper products, we'll type or write the headings that we decided on into the first row. Some of us might be interested in entering these into a database program, which has more organizational options. However, databases can be complicated, so we'll explore those in the future and stick with spreadsheets or hand-written tables.

Adding our headings

So we'll put the first box from the form, **SITE**, into the first column, first row; followed by **YEAR** in the next column, same row; then **AREA**, and then **SECTOR**.

However, when we get to **ELEVATION** we'll need to pause.

Within **ELEVATION** we have two pieces of information, Min: and Max:. As these both have to do with elevation, it'll be better to have these as two columns. We'll call these **ELEVATION_MIN** and **ELEVATION_MAX**.

Then we get to **STRATIGRAPHICAL UNIT**. This also has two pieces of information, but they are structured differently. Rather than both being numeric, it looks like there's a section that's usually a number and then a section that has specific options.

We'll also note that we have to break this into two boxes. The first is **STRATIGRAPHICAL UNIT NUMERIC** (for now, we may want to alter that title after looking at other context sheets) and **STRATIGRAPHICAL UNIT TYPE**, which will capture the checkboxes within that section.

As we've already seen, **DESCRIPTION** will probably need to be broken into separate boxes as well because it has spaces for both "Position within sector" area and a "Shape" area.

Deciding each headings data type

Congratulations, we've made some important decisions regarding how we're going to structure the data collation for this site! Deciding to separate boxes in the form into different column headings can be tough because, on the one hand, fewer columns might make it faster to enter things. On the other hand, more detail can be important for seeing patterns.

Once we've decided what headings we want, we'll need to name them in the table. Typically, we'll use the names that match the form. But for additional headings we'll want some consistency. So first write or type out the heading names into the place where we'll be collecting tabular data.

	B	C	D	E	F	G	H
1	Site	Area	Sector	Elevation_Min	Elevation_Max	Stratigraphical Unit_type	StratigraphicalUnit_number

The first row of our table should look something like this as we add our headings to it.

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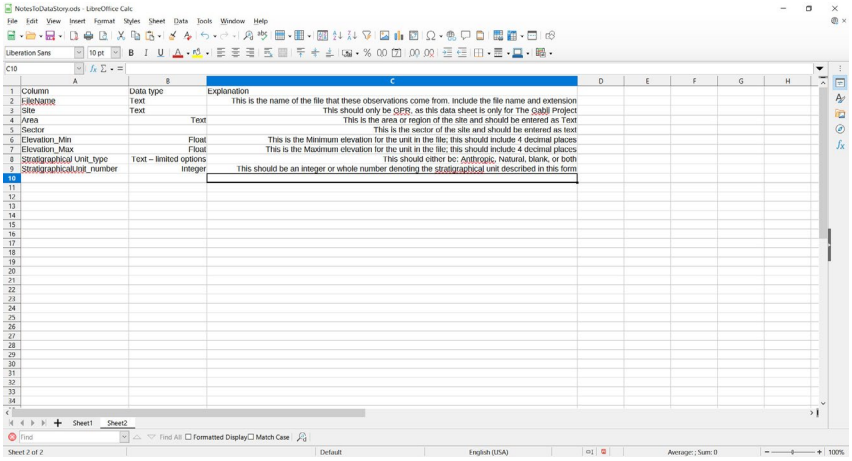
For headings we're adding, we should makes sure each version references the main box that the heading comes from, such as including "Description" as part of the title. For completely new headings, we'll just want to keep their formatting similar. Furthermore, because we're not using a database, we'll need to be careful and explicit about the kinds of information we'll want in the table.

To do this, we will add a README sheet that explains what should be entered under each column. These can be very useful if multiple people are doing data entry or if there is likely to be large gaps between data entry events.



Column	Data type	Explanation
Filename	Text	This is the name of the file that these observations come from. Include the file name and extension
Site	Text	This should only be GPR, as this data sheet is only for The Gabii Project
Area	Text	This is the area or region of the site and should be entered as Text
Sector		This is the sector of the site and should be entered as text
Elevation_Min	Float	This is the Minimum elevation for the unit in the file; this should include 4 decimal places
Elevation_Max	Float	This is the Maximum elevation for the unit in the file; this should include 4 decimal places
Stratigraphical Unit_type	Text – limited options	This should either be: Anthropic, Natural, blank, or both
StratigraphicalUnit_number	Integer	This should be an integer or whole number denoting the stratigraphical unit described in this form

A README sheet will look something like this if typed up in LibreOffice Calc.

However, this isn't exactly what we're working on right now so we'll explore the importance of READMEs in another tutorial.

Ok, back to the note-able parts. Once we've selected our headings, to remain consistent with how we enter data we should think about something called "data types". These are how those data are stored in the computer. To decide our data types we first need to consider whether the observation is a number or not a number. Is it 2, B (in separate boxes, a number and not a number) or not 2B (in one box, not a number).

Once we've decided that, there are different directions we'll need to go. Let's use the **ELEVATION** box as an example.

Is the information contained in that box a number?
- If no, it will be text, for now.
- If yes, is it a whole number or one with a decimal point?
  - If it's a whole number, it's an integer.
  - If it has a decimal point, it's a float.

There are other data types but we'll stick to these so that if we want to do some exploratory data analysis in our spreadsheet, all the data types within a heading or column are the same. Otherwise, we might try to find the average of 1 and fish and that just won't go anywhere.

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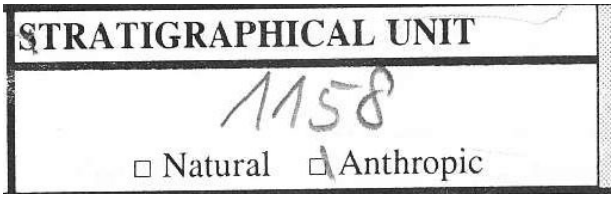
A basic rule is that if an observation is a number, use a numeral to symbolize it. For example, in the **STRATIGRAPHICAL UNIT** box, although we might not know exactly what that is, by taking a look at other context sheets we realized that this is always recorded as a number.

If they're whole numbers, they'll be integers. If they aren't whole numbers, meaning there's a decimal point or comma, then we'll call them floats for now. Floats are a more complicated kind of number that take up more storage in our computer but that's something we'll discuss in a different tutorial.



This is what the Stratigraphical unit box looks like.

Looking at the **STRATIGRAPHICAL UNIT** box, we might want to be more strict than typing up the observations as text or numerals. In this box there are a couple of words with little boxes next to them. Specifically, those are check-boxes and they suggest that there are a limited number of options for what can be recorded.

So we'll want to standardize how we enter those two options into our table. Some programs allow us to use check-boxes in our tables. More commonly though this is done by making sure we only use certain values during data entry. The first step is to decide how many options there actually are.

This will depend on if the options are mutually exclusive, meaning that only one state or the other can exist at the same time, if it's possible for both states to be true simultaneously, and if the possibility exists for neither state to occur. If all of these options are possible, then something that only has a two check box option on a form will actually have four possible ways to be entered. Neither option entered, one or the other option entered, or both options entered.

Specific data headings

So we know what headings we're entering:

SITE
YEAR
AREA
SECTOR
ELEVATION_MIN
ELEVATION_MAX
STRATIGRAPHICAL UNIT NUMERIC
STRATIGRAPHICAL UNIT TYPE
DESCRIPTION_PositionWithinSector
DESCRIPTION_Shape

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Based on our context sheet, what data types make sense for each of these? It may be good to look at a couple of sheets before deciding, but think about the questions we had for each box on the context sheets. Also since we're thinking about these and started a "README" sheet, we can put our thoughts into that table as well to keep ourselves organized.

Here are some of our explorations of the headings we're using for the exercise. Some of these may have multiple options but for now stick with these as we develop these skills.

SITE looks like it's printed directly onto the sheet and is always "GPR". As this is alphabetical, we'll want this to always be a text value. However, more importantly, this value doesn't change. We'll talk about how do deal with that in the next section.

YEAR looks like it is the space where we put the year of the project. If we look at the context sheets, some have two digits and others have four digits. However, they're always whole numbers so these are probably integers. As we enter these data, we should standardize these to either two digits or four digits. For now, go with four digits, even if the paperwork might only have two. While we're taking some liberties, as this isn't *exactly* what is in the form, it will ensure there isn't confusion over whether excavations happened in 2010 or 1910 or even 10AD depending on how long they've been excavating that site.

AREA looks like it is the space where we are putting what named part of the site the context sheet is from. However, it is not the area, as in the extent of it, such as by multiplying the width times the height. Instead, this area is a named region of the site. It often seems to be just a single letter, so we'll make sure it only accepts text values and choose to always enter those as capital letters so we don't confuse lower case L's and uppercase I's.

SECTOR looks like it's often blank. We won't make any decisions about that for now as it's unclear how recorders were capturing that information. We'll talk about blanks a little later on.

ELEVATION_MIN looks like it is the space where we put the minimum elevation for the context. It looks like this is a measurement taken with particular units. It looks like it has a number of decimal points past the original number but the number past the decimal varies. This should be a numeric value then and typically one that has decimal places, so a float.

ELEVATION_MAX looks like it is the space where we put the maximum elevation for the context rather than the minimum. Because it matches with the minimum, we want to capture this just like we did for **ELEVATION_MIN**.

We'll pause for a second here. It looks like the two values that go into the **ELEVATION** headings are measurements. This means that they should probably have units, for example millimeters (mm) or centimeters (cm), denoting how much of what was measured.

It does not look like these forms specify those. The project took place in Italy though, so it's likely that the units are the International System of units or SI units. We may consider adding a heading called **ELEVATION_UNITS**, to specify this but without further information we'll leave our headings as they are.

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STRATIGRAPHICAL_UNIT_NUMERIC looks like it is the space where we put a number for the stratigraphical unit. These appear to be typically whole numbers, or integers. So we want to make sure that no decimal points are included in this box when we enter it.

STRATIGRAPHICAL_UNIT_TYPE looks like two check boxes, one for "Natural" and one for "Anthropic". These probably describe something about the stratigraphy and whether it occurred with or without human influence. Based on this, it's unlikely that a single unit would be both natural and anthropic. However, it is possible that this section could be left blank, so we want to make sure we are aware of that. Because we're working with basic data types this will be a text field but there should only be three potential values for that heading in the table.

DESCRIPTION_PositionWithinSector looks like it describes where in the sector the context recorded was. While they all have something about direction, such as north or south, what this is measured in relation to, changes. Because of that, they are not standardized enough for separate headings. So unfortunately, we'll leave it as one of the dreaded free-form text options.

DESCRIPTION_Shape looks like it describes the shape of the context recorded. These should be one word descriptions. While this will also be a text field, it should be limited to only words associated with potential shapes. However, sometimes there's a little more to that description therefore, we should add an additional heading.

DESCRIPTION_Shape_Notes does not exist as its own box on the form. However, some context sheets have both the named shape and comment on that shape with further observations. This area also sometimes includes observations that are not always obviously relevant to shape but should be captured anyways. While the possible shapes are set, this section should be another free-text option to account for what folks noted regarding the shape of the context and other possible information captured in the **DESCRIPTION** box.

Headings without a space on the form

Besides adding headings that break up the sections of the form into their own separate fields, it may be important to add headings to the table that fill in information not covered in the form. Most often, these will be headings that describe the process of data entry. However, they can also be things that were left off, like the lack of units regarding the elevation section that we noticed.

An important one, that may have a form-based counterpart, is a Notes heading. While in some cases the bane of data-entry standardization, "Notes" can be important for capturing observations not described elsewhere. Notes can both exist as a space on a form that the person doing data entry types up and Notes made by the person doing data entry. These can be separated into headings such as "FieldNotes" for those that are transcribed from paperwork and "LabNotes" to account for observations made during data entry.

Other potential headings might be entered by that will allow the person entering the data to note who did the work of entering the data in case someone wants to know more about the process and decisions the transcriber made for how to fill in the data.

For this tutorial, we'll add a LabNotes heading and use that for exploring what happens when we encounter blanks in the context sheet.

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Filling in the table

Now that we've decided the structure of the table, we can go ahead and put the information that we wanted to gather into it. As a test to see if that structure makes sense, go back to the beginning of the tutorial and download a few more context sheets. That way we can see if how we organized the data makes sense or if we'll need to edit it in some fashion.

Or, instead of repeating those steps, we can just click this link, so that we can download more context sheets to enter.

We should do this a number of times so that we evaluate the data structure we've created and so that we have enough observations to answer some archaeological questions. For the purposes of this tutorial, let's try to input at least ten context sheets. It's a lot to type but ten entries should allow us to test the structure of our table to make sure it does what we need it to.

However, it's also few enough entries that if our structure doesn't work, redoing those won't feel like a waste of time. Better to change it at ten entries than 10,000 entries because something significant was missed or entered incorrectly.

This vetting also reminds us that sometimes our data entry structure just doesn't work in practice. Or something comes up that requires us to re-evaluate that structure and that's fine. Likely too is that we'll need to revisit our data structure as new questions develop as data entry goes on. However, for any changes we make we should take care to describe them so future researchers understand what changed and why.

How to fill in the table

We discussed data types previously, and now that we're actually entering data we suggest getting some headphones and listening to music or podcasts during this portion! Remember, we want to try to keep what we actually enter as standardized as possible.

On that note: copy and paste is our friend. Copy and paste might, in fact, be our best friend. The friend who keeps the notes we wrote them in high school and cherishes them forever. Or burns them because they know if those ever got out no one would respect our work as archaeologists. The author of this tutorial is definitely not writing from personal experience. (Editor's Note: Neither is the editor.)

That aside, for any fields that contain text, using copy and paste to ensure consistency is important.

Similarly, we want to make sure that our numbers are entered the same way, and that we don't do such things as, um, enter numbers as text. This is something we can set in the formatting section of a column in many spreadsheet programs.

Particularly for the section labeled "Elevation". Based on the forms, this was consistently measured to multiple digits past the decimal place. Unfortunately, sometimes it's three places and sometimes it's four. While this was probably a standard that should have been set in the field, we're going to choose to capture the most detail. Therefore, we might set the formatting to always show us four places past the decimal. This will add an ending zero, possibly implying more precision than was in the field but it also means we're not rounding when someone in the field put in a specific value in that decimal place.

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Dealing with "blanks"

One of the great and terrible aspects of archaeological data entry is the blank form or blank box. We open up or take a look at a particular set of paperwork, whether it was written yesterday or 20 years ago, and we look at a specific area, which might have the information we really want, and we see that that area has…nothing in it.

We may think a number of things at this moment. Such as:

- How did I get here?
- What happened to my beautiful form?
- Is this even my site?
- Why did we ask this particular question?
- Does this blank mean that this didn't exist?
- Did this question not apply to this particular observation?
- And while we hope this isn't the case but sometimes it is, did the excavator or note taker simply forget to fill in this section of the form?

We are human after all and accidents happen. However, a blank section in a form does not necessarily tell us which of these options is the truth.

Hopefully, there's someone we can chat with for clarification on this. But sometimes there isn't, and we need to decide for ourselves what, if any, information we take from the fact that something was *not* recorded.

Someone thought that it was an important box to have on the form. Yet, for this particular context sheet it was crossed out or left blank. So what do we choose to put down in this space within our growing table? Did we plan for this possibility in our data structure? And how do we, as the, er, data enter-ers signal the difference between the previously discussed options?

There are multiple ways to do this and one method may outweigh another depending on the project. For this tutorial though, we're going to make note of any blank or crossed out fields in the "LabNotes" heading we created earlier.

Under that heading, we can put any notes that we thought of while entering the observations, explain why any of the attributes were left blank, and why if we know. It may also be valuable to note either in that heading or in the "README" sheet that any unexplained blanks are data enter-er error.

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A tabular dataset

After a few observations, take stock and see if we can pick out any particular patterns in the context sheets that we collected. If using a spreadsheet program, we can try filtering the data to see if there's something that we can say about the site based on the information we've put into the table. If we're unfamiliar with filtering in our spreadsheet program, we can check out the spreadsheet tutorial section of *Cow-culatating Your Data with Spreadsheets and R*.

- Are there any particular words that get repeated in particular fields?
- Do some elevations have more notes about them than others?

Now what?

Congratulations! We've made, or are in the process of, making our very own data set from the resources shared by The Gabii Project. This means that we made some data entry decisions that created a structure for our data. It's not the largest data set out there, but hopefully we've recorded enough observations to feel like we understand how to turn notes into data. And if we aren't comfortable doing data entry yet, there are 180+ context sheets we can use to keep practicing.

Once a few are entered, we've explored what's there, and looked for patterns, we will probably want to do more with these data. Why? Well this is only the first part of the cycle. We've turned our structured notes into data. But we don't really want data, we want information and, even beyond that, knowledge.

To get to that, we'll need to move to the next section of this tutorial. Specifically, if we entered in the information from the **DESCRIPTION** section for 13675 there were some belongings, sometimes called artifacts, listed in that section. We'll focus on the information about those specific belongings and notes from other sections of that context sheet as we transition from data to narrative to say something about the site.

Key concepts used in this tutorial

- data structure - how we choose to give form or organize our data
- paperwork or documentation - the notes and ways we choose record our observations of the archaeological record
- columns, attributes, headings - the specific observations we want to take note of
- entries, rows, observations - these refer to the specific thing we are observing and the particular instance of the attributes it had
- data types - the ways that information are stored in a digital environment in a standardized fashion.

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Welcome to Part Two, Data to Narrative!

In Part Two of this tutorial we will learn to:

1. Utilize data from tables and context sheets to create data narratives
2. Analyze existing data narratives for theoretical approach and exclusionary factors
3. Apply a narrative lens to a data narrative of our your own creation

Key concepts used in this tutorial

- data narrative
- narrative lens
- hegemonic storytelling
- bias
- tone
- jargon
- audience
- language level

Introduction

We hope you've taken a few minutes to relax after the last portion of the tutorial. There was a lot to do there. The process of turning notes into data is a crucial part of building knowledge, but most of the time, it's not the last step. So let's talk about what we do, when we're done doing data.

The goal of every archaeological project is, ultimately, to introduce more knowledge about the past to the people of the present and future. If we don't find a way to make the data we've carefully collected, digitized, and analyzed available and understandable to the public, what we did wasn't science; it was just digging holes and making spreadsheets. Which brings us to the second part of this tutorial: turning data into narrative.

Humans are storytellers. We've always been storytellers. Every culture, from modern day to those of the earliest paintings in caves, has told stories. People have always looked at the world around them, considered their place in it, and tried to share what they know, value, and regard as important.

As archaeologists, we occupy a special place in this long chain of storytellers, because not only do we tell the stories of our own lives and cultures, but through our work, we retell the stories of peoples in the past. This allows a connection with the past that is rare, and special, and with that privilege, we have a responsibility to share the stories we learn with others.

Take a moment and sit with that. Think about what that responsibility means.

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All of these are concepts or issues that archaeologists must consider in their practice.

A data narrative is a particular type of story that is grounded in scientifically derived findings and geared towards an audience. That audience may be very big, or it may only be a few people. It may be adults, or children, or both.

For the most part though, the audience won't be other archaeologists. Probably, it won't be other data scientists either. So while we can look together at our digitized tabular data and see things in it, that's not the format that most people find engaging when they're learning about the past.

We have to go from data back to narrative. We have to tell them a story.

What data will we be working with?

The story we're going to be telling today is about a particular part of a particular archaeological site in a particular part of present-day Italy. We're going to be using data from the same site that we worked with in the other part of this tutorial, Gabii. To make things a bit easier though, we've limited this part of the tutorial to only information that can be reached off of this page.

Through this curated collection of data, which contains tabular data, images of artifacts, context sheets, and maps, we will tell stories about the people of Gabii. Take a few minutes to look through the files and download any that seem interesting to you. Maybe reorganize them?

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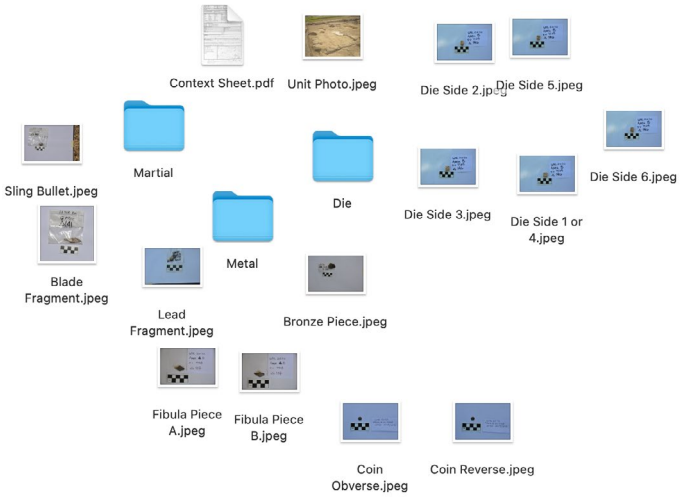
Below is a screenshot of one way we could organize these files, but everyone may have their own way of shuffling things to make sense of them. There's no one way to organize this information, which brings us to considering point of view. Or, what we call it when designing data narratives, our narrative lens.



We chose to download and rename images of the 'special finds'. They're organized by association, but could easily be regrouped in many different ways.

What is a narrative lens?

In an everyday story, the narrative lens is what we use to see, and understand, the story that we're experiencing. Think of it as a perspective. All stories are written with a perspective, whether their authors intend to do so or not.

As scientists writing data narratives, we have to be very conscious of our perspective, because remember, we're telling a story that is based on data. The perspective we put on the data narrative we create will influence how that data are perceived, and understood, by our audience.

Let's look at a few different narrative lenses that we may employ when we're creating our data narratives. We're going to look at what makes a narrative standardized, or 'hegemonic', at feminist narratives, at queered narratives, and at meta-archaeological narratives. If those four lenses sound familiar, congratulations, there was a good theory teacher involved somewhere along the line!

Hegemonic narratives

The hegemonic narrative in archaeology is the standard, basic, probably heard it in a million formats, story about the past. A hegemonic narrative is the baseline of archaeological storytelling. In hegemonic narratives, the archaeological story is distilled down. These narratives, though basic and repetitive, can be useful to establish what the tropes are in archaeological narratives of a particular culture, period, and place. It's important to note that hegemonic narratives aren't necessarily wrong, sometimes the standard story is the story. We just want to be able to share narratives that consider multiple possibilities.

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Common hegemonic narratives in archaeology include: stories about the Roman Empire told from the perspective of a soldier or gladiator, stories about the medieval period told from the perspective of a high ranking member of the nobility, and stories about Native Americans told from the perspective of a shaman.

Feminist narratives

An approach from feminist archaeology asks the reader to consider context, the lack of neutrality in data. It asks the reader to rethink hierarchies of gender and access to material goods within the archaeological record. Feminist narratives don't necessarily have a female central figure, but do more often center peoples of the past who are less typically represented in archaeological narratives, like children, the elderly, the dis/abled, and others who might be marginalized through cultural factors.

An example of a feminist narrative in archaeology would include: stories about the Roman Empire told from the perspective of a child in an occupied territory, stories about the medieval period told from the perspective of a shepherd's wife, and stories about Native Americans told from the perspective of a still existing pre-contact oral tradition.

Queered narratives

In queered narratives we consider the presence of artifacts from a perspective outside of their past role as commodities and their present role as proxies for the past. We consider the concept of 'difference', and 'indifference', and illustrate how space and place are fluid, and how power attached to place is fluid. Narratives with this lens may focus on the archaeological unit as a transitional space, a place that once had definition, but is currently in a liminal state.

An example of a queered narrative in archaeology would include: stories about the Roman Empire told from the perspective of a non-Roman living outside of the empire, stories about the medieval period told from the perspective of a Jew in England, and stories about Native Americans told from the perspective of a modern displaced nation.

Meta narratives

Finally, the fourth approach we're presenting is a meta-archaeological narrative. This is where open data sets like this one from Gabii are great to work with. We're able to interact with digital copies of first-line interactions with artifacts, units, and sites, through scanned copies of site forms, like we looked at before in this tutorial! So we can get details on what the process of excavation was like—linked data lets us look beyond the site form to compare the time of excavation to conditions on the day, and lets us look at the role of a single excavator across a site, or across multiple sites. Meta-narratives take a specific experience and universalize it for the discipline.

An example of a meta narrative in archaeology would include: stories about the Roman Empire told from the perspective of a student attending their first field school in Italy, stories about the medieval period told from the perspective of a modern potter recreating medieval ceramics, and stories about Native Americans told from the perspective of a Native archaeologist working in cultural resource management in the United States.

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Writing a narrative

So are we ready to try writing a data narrative? We'll work on this together, so that you don't have to start from scratch.

Consider the following narrative.

> The smell of cooking meat and the stink of other people's bodies was inescapable. Crouching on the uneven floor, Aemilianus shook the die, blowing on it gently. He rattled it out of his hand. It rolled, and rolled, and rolled -- there was the six, and then, and then, it tipped, coming to a stop.
>
> A two. He couldn't win on a two.
>
> Reaching into the pouch affixed to the mail of his armor, he pulled out a few thin bronze coins, flipping them to the winner. Someone called his name, and he looked up. Time to go.
>
> He left the die. It wasn't doing him any favors.

Let's look at each part of this narrative, and see where it came from, and what purpose it served. Here's the text, again, this time highlighted. Each highlight corresponds to a piece of data we have within our curated dataset.

> The smell of cooking meat and the stink of other people's bodies was inescapable. Crouching on the uneven floor, Aemilianus shook the die, blowing on it gently. He rattled it out of his hand. It rolled, and rolled, and rolled -- there was the six, and then, and then, it tipped, coming to a stop.
>
> A two. He couldn't win on a two.
>
> Reaching into the pouch affixed to the mail of his armor, he pulled out a few thin bronze coins, flipping them to the winner. Someone called his name, and he looked up. Time to go.
>
> He left the die. It wasn't doing him any favors.

Have you found the origin of each piece of data? Remember, the data might be an artifact, it might be a unit description, it might be a soil sample, or it might be pure measurements. What kind of narrative, based on the types we talked about before, is this?

The first highlighted section refers to three samples of faunal materials.
The second highlighted section refers to the shape and layer surface.
The third highlighted section refers to a gaming die listed as a special find.
The fourth highlighted section refers to a bronze coin listed as a special find.

This is a hegemonic narrative. This is a typical Roman story. We might see a narrative like this in film or television, in a novel, on a sign at a site, or in a museum booklet.

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In this narrative, the protagonist is a man. He's in the military. His life is blood, and sweat, and drink, and food, and though he isn't within the white-robed elite, he has enough money that he can gamble without worrying about a loss. He exists within a larger military structure that tropes in media representation of Romans makes easy to allude to.

Now it's your turn. Pick three pieces of evidence from your curated dataset. Take about five minutes and write a one-paragraph story, incorporating your three pieces of data. Start with a hegemonic lens, we don't have to get fancy and show off our theory...yet.

If you're worried about how to get started, consider these ideas:

Who are you telling this story to?
What is the most important point you want to make?
What stories have you heard in the past that might make good foundations for this story?

It's okay to be outlandish in the first draft. Maybe it would help to just write dialogue, like a conversation? Maybe it would help to draw stick figures or storyboards of what happens? Maybe going online and looking up names from the period will spark an idea about the narrator?

Analysing your narrative

A critical aspect of creating data narratives is reflection. Gabii wasn't built in a day, and neither are perfectly crafted narratives! We're going to look at the narrative you've written now, and consider how it's effective, and where we might want to change things up.

Looking for bias

The first thing we should look for is bias. Before getting worried, it's normal to find some bias in our narratives! This is that thing we talked about before, the writer's perspective. What we want to do is make sure that our own perspective as archaeologists, data scientists, and writers isn't leading our readers to incorrect assumptions about our data.

If you haven't tried out our *Cow-culating Your Data with Spreadsheets and R* tutorial, you might consider giving it a go after this. In that tutorial, we talk in more detail about types of data bias, and how interpretations can be skewed.

For now though, let's look at each piece of data incorporated into the narrative we started with. For each piece of data, answer the following questions.

Why did I choose this piece of data over another?
Why is this piece of data important to share with the public?
Is this piece of data representative of, or contrary to, the time period/site/unit I want to share?

Now have a look at the data you used in the narrative you created. Answer the same questions. If you find that the answers aren't what you want or what you expected, go ahead and make new choices.

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Looking for tone

The second thing we should look for is our tone. No one is going to ask us to sing, don't worry! When we're looking at our tone in data narratives, we mean we're looking at how our word choices and sentence structure creates a 'feeling' within the narrative.

In the hegemonic narrative we started with, the tone is intended to reflect the physicality of the main character's life. Most of what the narrative shares is him 'doing' something, or encountering something sensorial. What does he smell? What does he see? What does he do?

Have a look at the narrative we started with.

> The smell of cooking meat and the stink of other people's bodies was inescapable. Crouching on the uneven floor, Aemilianus shook the die, blowing on it gently. He rattled it out of his hand. It rolled, and rolled, and rolled -- there was the six, and then, and then, it tipped, coming to a stop.
>
> A two. He couldn't win on a two.
>
> Reaching into the pouch affixed to the mail of his armor, he pulled out a few thin bronze coins, flipping them to the winner. Someone called his name, and he looked up. Time to go.
>
> He left the die. It wasn't doing him any favors.

A thin bronze disc, which may have been used as small change currency.

Can you identify the tone?
What writing choices support this tone?
Does this intended tone convey the data?

Now have a look at the narrative you created. Answer the same questions.

If the tone isn't what was intended, take a few minutes to try changing some of the word choices. Give the story a little more attention. Editing is our friend if we want to create an engaging story.

Looking for jargon

The third thing we should look for is jargon. Jargon is language or terms used by those within a profession that is so specific to that profession that it's rarely used or understood by those outside of the field. Archaeologists love jargon. We throw around Munsell colors. We handily discuss areas, units, and features. Sometimes we mention sherds, but other times we mention shards. We can write for pages and pages about stratigraphy.

In a report, or an article, this jargon can be necessary. In those formats we're speaking to other archaeologists, and we use the jargon of our discipline to share information with one another, because it's expected that we'll all be familiar with the terms. But outside of professional communication, the rest of the world doesn't necessarily know our lingo.

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Have a look at the narrative we started with.

> The smell of cooking meat and the stink of other people's bodies was inescapable. Crouching on the uneven floor, Aemilianus shook the die, blowing on it gently. He rattled it out of his hand. It rolled, and rolled, and rolled -- there was the six, and then, and then, it tipped, coming to a stop.
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> Reaching into the pouch affixed to the mail of his armor, he pulled out a few thin bronze coins, flipping them to the winner. Someone called his name, and he looked up. Time to go.
>
> He left the die. It wasn't doing him any favors.

Are there any terms that are mostly common within archaeology?
Are there any terms that are too specific?
Are there any terms that might pull the reader 'out' of the narrative?

Now have a look at the narrative you created. Answer the same questions.

If you find any jargon in your narrative, take a few minutes to replace those terms with more audience-friendly choices. Give your draft another polish.

Knowing your audience

And now, here we are, with a solid draft that does what we want it to do, and reads how we want it to read. But oh, we forgot something...we have to consider our audience.

We should always know, before we write, who our intended audience is. Once we have that information, there are two things to consider to better tailor a data narrative to that audience.

Language proficiency

The ACTFL, the American Council on the Teaching of Foreign Languages, assess proficiency as what a language user can do regardless of where, when or how a language was acquired. Because the audience for archaeological content can vary so widely, we have to take into account that not all of our readers will have the same level of proficiency. Remember when we talked about jargon? Jargon is extremely difficult for those with lower levels of language proficiency to understand.

When we're writing for an audience, we scale our assumed level of language proficiency down. Even if we're writing for what we assume is a college-level educated public, we scale the language we use down. There are lots of tools and aids available online to help with checking our language levels. Three of the most common formulas for checking language and reading levels are:

The Flesch Reading Ease
The Flesch-Kincaid Readability Test
The Gunning Fog Index

Calculators for these formulas, and others, are available online.

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Length

The other thing we have to consider is the length of what we've written. Think about the experience as a whole. Is this piece of writing the only reading the audience will be encountering? If so, we can get away with writing a bit more. If our piece is just one segment of a larger written encounter, we need to consider scaling down what's presented.

Readers are more likely to engage with the content if it's presented in bite-sized segments. Furthermore, they're more likely to engage with more segments if those segments are themselves short.

A good best practice is no more than 100 words for anything that's going up as a sign, or placard, and no more than 250 words if it's being presented as a hand-held written supplement. If we're presenting our work as a blog post, or part of a website, 500 words is the max!

The hegemonic narrative we started with is 107 words. Let's take a look at the narrative you wrote.

How many words long is it?
How many sentences does it have?
What's the reading level?
Was it possible to stick to, roughly, just one paragraph?

Rewriting the narrative

Now that we've spent time with our hegemonic narrative, and we've got it: using solid data, written in an appropriate tone, devoid of jargon, set to our chosen language level, and of a good length...it's time to apply a different narrative lens!

For each of the narrative lenses we talked about earlier, we're going to use the same process that we initially used to write our hegemonic narrative. We're going to use the same three pieces of data that we used before, too.

The goal here is to think differently about the same pieces of information. In archaeology, 'theory' is the intellectual or philosophical framework that archaeologists apply to make their interpretations of the past. There are lots of different archaeological theories, and theoretical perspectives have changed over time. Here, we'll consider how several theoretical perspectives, aka, narrative lenses, influence the stories we tell about the past.

We may find, as we look at each narrative lens, that we've naturally already fallen into writing from one of these theoretical perspectives. If so, try to identify what about the data we highlighted led to that narrative lens.

Are there other pieces of data within the curated dataset that might reinforce that finding?
Are there pieces of data that refute it?
How could we rewrite the data narrative to account for the other data sources?

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Rewriting through a lens from feminist archaeology

Remember, when we're writing from a perspective in feminist archaeology, think gender, hierarchies, power, and access.

Here's an example of the narrative we shared earlier, this time written with a feminist archaeological lens.

> It was her turn, but he kept hoarding the die. He hoarded the thin bronze disc, too, and all of the bits of metal they'd scrounged and hidden here, in their secret place.
>
> Helvia and Helvius were meant to be minding the goats, but the goats mostly minded themselves. There was time to play, and this falling down room, with its uneven floor, made the perfect place to hide away and pretend.
>
> She toed a broken piece of thick-walled pottery and waited for her brother to remember it was her turn.

A 'leech style' fibula pin.

This narrative includes the same artifacts and the same placements within the implied area, but we replace the tropes of hegemonic narration with actors and goals out of feminist archaeological practice.

Instead of a militarized male, an identity associated with access to social and economic power, we challenge unequal power structures by considering a view of the site from the position of children. Specifically, the narrative is from the perspective of a female child, but that perspective is viewed in its context of power alongside a male child of similar age and sociocultural status.

Looking at your rewrite, consider:

Are hegemonic archaeological norms of gender, power, or control subverted in this writing?
Does this writing reflect a view on the past through an uncommonly represented narrator?
Does the data used as evidence for the narrative still apply?

This plan shows the excavated areas at Gabii. Area B, which this tutorial uses data from, is roughly in the center. Image originally published in A Mid-Republican House From Gabii, available under CC-BY SA.

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Rewriting through a lens from queer archaeology

Remember, when we're writing from a perspective in queer archaeology, to think not only about sexuality, but also more broadly about commodification, liminality, and artifacts outside of functioning as proxies for the past.

Here's an example of the narrative we shared earlier, written with a queer archaeological lens.

> He threw a shovelful of dirt on the floor, stamped on it, tamped it down. He threw another shovelful, and again, and again, building the uneven surface to something more usable, something he could live on. If he didn't get this space sorted, he wouldn't have anywhere else to go. This space was free and available.
>
> Something white stuck out against the dust and soil. Adamo bent down, using the tip of his finger to flick the loose covering (not tamped enough, there was more work to do) off of a bone-colored square.
>
> A broken die. Pips where the four should be were gone. Worthless. The past had left behind so much junk here. He dropped the die and threw another shovelful of dirt.

Dice. Double threes ftw!

This narrative includes the same artifacts and the same placements within the implied area, but we replace the tropes of hegemonic narration with actors and goals out of queer archaeological practice.Though our protagonist is gendered male, he's positioned as an outsider, and his socioeconomic power is curtailed. This narrative focuses more on the archaeological unit as a space, instead of as a place in time.

Looking at your rewrite, consider:

Are hegemonic narratives of sexuality, power, privilege or space subverted in this narrative?
Does this writing reflect a view on the past through a queer or gender diverse narrator?
Does the data used as evidence for the narrative still apply?

This is Area B at the site of Gabii. The remains of walls can be seen, as can several cuts in the floor. One of these cuts held Grave 18, and the other held an 'olla', a pottery vessel.

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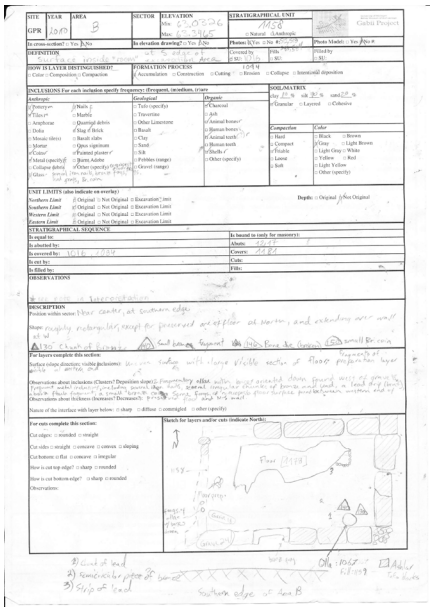
Rewriting through a meta-archaeological lens

Writing a meta-perspective in data narratives can be fun. It gives the public a little peek behind the curtain of how archaeology happens, and it lets us incorporate our own experiences, albeit generalized, into a depiction of our discipline.



A context sheet for Area B, Unit 1158.

It was so hot. It was hot all the time. Standing at the edge of their unit -- the ground they had revealed as a small room with an uneven floor, they considered their day's labor, and they sweated.

The whole day, spent sweating and dusty, and it would probably end up in the report as just another post-abandonment surface, if it made it in the report at all. Their finds bags had some bones and charcoal, a bronze coin, a few bits of metal, a broken fibula pin, and a compact bone die, the four-side of its matching pips missing. There was a time they'd have had all of these things on their 'to find' bucket list.

Time for a Peroni. They signed their name on the context sheet and stretched.

Looking at the context sheet for the day this unit was excavated, we can cross reference with historical weather data to find out conditions on the day. The day this unit was excavated was the 13th of July, 2010. The temperature ranged from 28 to 32 degrees Celsius, or 84 to 89 degrees Fahrenheit. It was a hot day in July just outside of Rome.

This context sheet was one of multiple context sheets completed by the same person, and we can infer the amount of effort this unit took by the detail and attention to detail on the form. We're not judging their enthusiasm or work ethic, we just know what it's like to be at the end of that day! The narrative we're telling through this final approach is one that most archaeologists can recognize, regardless of site location, but one that doesn't frequently occur in mass media—it's the end of the day, the archaeologist is tired, they're feeling a little jaded. It's taking a specific experience and universalizing it for the discipline.

Looking at your rewrite, consider:

Are hegemonic narratives of archaeologists as explorers, adventurers, and collectors subverted in this narrative?
Does this writing reflect a view on the past through the perspective of an archaeologist or scientist as narrator?
Does the data used as evidence for the narrative still apply?

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The Published Narrative

Now, having looked at multiple short narratives created by the Digital Data Stories team, and having written your own narratives, it's worth having a look at the 'official' narrative, as presented by the Gabii Project team in their publication on the site, *A Mid-Republican House From Gabii*. Though this narrative draws on more data than we worked with in this tutorial, hopefully in reading it you can now pick out some of the things we talked about. This text is presented verbatim, thanks to a generous CC-BY SA License.

As you read the project's narrative, consider:

What data is present in this narrative?
What is the narrative's tone?
Can you identify any bias in the narrative?
Does the narrative include any jargon?
What is the general reading level of the narrative?
Who is the narrative's intended audience?
What narrative lens is the narrative told through?

The town of Gabii is shaped like the flattened skeleton of a fish, a long curved spine with ribs splayed out, running off either side. It sits on the southern slopes of a low conical mound with a crater at its center; a bit of landscape whose topography was defined by a long-dormant volcano. The town's main east-west thoroughfare, the Via Gabina, traces the contour of the slopes of the crater, not far from the base of the mound, linking Rome to Praeneste. Between this road and the rim of the crater that defines the northern limit of the town, a series of terraces rise up. South of the road, the town continues across more level terrain.

Once upon a time in this town, there was a house. In fact, there were a number of houses, but this story is about a particular house: the Tincu House. This house was built two terraces north of the main road and one block east of a particularly important side street that runs down the slope from the crater's rim to join up with the Via Gabina. The location of this house, tucked just off this crossroad, gives shape to its story.

To tell the story of this house, we must begin before the house was built and even before the streets that defined the limits of the property were laid out or the terraces were cut into the bedrock. At this time, huts, yards, and other small structures, some grouped into compounds and ringed with their own walls, were scattered across the slope of the crater. In the place where our house will be built, someone is living or working. Later events largely obscure the actions and lives of these early occupants, but we know they were there. We see a scrap of a floor—a compact clay surface, disconnected bits of stone-built architecture, and a few cuts in the bedrock that may have served as drains or as foundation trenches for walls. A thorough reorganization of the town in the late 5th or early 4th c. BCE disrupted whatever these people were doing. The decision was taken to lay out a network of streets that would divide the slope and the plain below into a series of long narrow blocks. Within each block, terraces delimited by rock-cut roads provided relatively flat areas on which new structures could be built. Our house is built in a space defined by a street on the west and terracing cuts on the north and south. The eastern limit of the property ends roughly halfway to the next street over, but it is not defined by the skeleton of streets and terraces that provide the physical frame for the town.

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After a pause when the block may have stood empty, the building of the house begins in the early 3rd c. BCE—roughly 280-260 BCE—with the serious business of creating a level and well-drained surface. We see a large cut into the bedrock to construct a level terrace, as well as the installation of channels cut into the bedrock, designed to funnel water south down the slope. Soil and rubble are spread across the building site where the house will appear. The initial outlines of the house are simple enough, a large rectangular courtyard flanked on its eastern side by four rooms. The courtyard is entered from the west through a doorway onto the north-south side street. The walled courtyard space was built open to the sky, and a well was set into the ground in the northern part. The rooms themselves are entered from the courtyard, and some may have connected to one another. The roofs slope outward, dumping rainwater into drains running along the exterior walls of the house. The house is built almost exclusively of local stone—the lapis Gabinus quarried just up the hill.

What happened in the house when it was in its first, simplest form? The usual things, we must suppose. People lived there, cooked and ate, argued and misplaced things, swept and trod down the floors.

Sooner or later, as often happens with houses and their occupants, the domestic arrangements proved unsatisfactory—not enough space/light or just things wearing down a bit. The floors in the eastern rooms are redone—scraped, leveled, re-packed, and resurfaced. A series of new rooms are added along the southern edge of the courtyard, providing more places to keep things, work, or have a bit of privacy. The house is still a house—just a nicer one with some extra rooms and new floors. Then the change comes.

What makes a family leave their home or allow it to be transformed into something that is not a home, even if they still own it? We can't say, but we know it happened here. The house stops being a house, and a complete transformation into a place with another purpose begins. Two large rooms are added on the northern end of the house, taking over part of what had been an open courtyard. The door through the wall on the western side of the courtyard is blocked up, and a new entrance to the courtyard is built from the south, fronting on a drive that branches off the north-south road that runs up from the Praenestina. A small vestibule is built between the new doorway and the courtyard. Further rooms, small and confining, are added on the southern side of the courtyard, and a large basin is installed in the courtyard, taking up more than its fair share of space. A new wall is added in the courtyard, running parallel to the line of the original rooms, blocking off the formerly easy access to them. The kinds of stones used to build the walls are changed, with the local lapis Gabinus swapped out for the lighter if less durable tufo lionato. The sewers and drains that used to collect water running off the eaves outside the house are closed off. The roofs are rearranged, and instead of water dumping outward and running away from the property, it is gathered inward, pouring down the sloping tiles into the courtyard, perhaps spouted into the new central basin.

Why do all this? Why invert the roofs, block the drains, install a basin, add walls, and close and open doorways. It's clear at this point that the house no longer works as a house. But it looks like a house, at least as seen from the interior of the courtyard or from the road looking across the threshold. There would be roofs, a basin to catch water, and some surrounding rooms—all the things one might expect. Like a stage set of a house, there's no home behind it, but the look is good. What's behind the facade?

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For a time, the rooms behind the courtyard remain in use, serving some unknown, backstage purpose. Offices, storage, or something along those lines seem likely. The spaces certainly aren't well lit, easily accessed, or prominently positioned. This transformation, from a house into a something else, seems to be associated with the arrival of new neighbors. The installation of a large public complex one block to the west doubtless affected the story of this house. Its presence seems the most likely explanation for the radical change witnessed here: the transformation from a house into an annex. This conversion from private domestic to "other use" space requires some explanation. To propose an answer, we must stretch the limits of imagination. Maybe this public complex needed an administrative area. Maybe the house lot was bought and transformed, effectively split into two parts. In the former courtyard, the new walls, rearranged roofs, and basin create the illusion of standing in the interior court of a modern house. Hiding behind the domestic facade are spaces that might work for official business and storage. Together, these spaces create the illusion of the public parts of a house, with functional, if inconvenient, spaces hidden from view.

Something like this must have happened. We can be certain there is a transformation from a place that acts as a house, with fairly modest and uncomplicated spatial arrangements and a look as much rural farmhouse as contemporary urban dwelling, into a place that looks like a more modern "Roman" house but doesn't act as one.

The activities that required the illusion of a house wind down or shift elsewhere. The rooms hidden behind the facade are repurposed again, as quarrying in this part of the city creeps south down the slopes. The northern wall of the house comes down, and dumps accumulate within some of the rooms, while others remain used for something that required a clear space in reasonable repair. Eventually, the whole place is given up.

The basin in the courtyard is removed, and rubble and dirt are dumped in fits and starts. Over time, this space fills with debris. The walls of the house come down; whether of their own accord or through intentional removal is uncertain. The useful material is carted off elsewhere, and more dirt accumulates. In the end, there is no longer a house, a facade of a house, or a rough repurposed industrial space. There is just an empty lot once more. Then the next thing happens.

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Concluding Thoughts

In this tutorial, you have learned to define a data narrative, to choose data to use in your narrative, to write a basic data narrative, and to apply a narrative lens (aka, a theoretical perspective) to your interpretation. You have considered how bias, tone, jargon, and audience can influence the choices you make in narrative construction, and you have analyzed both a provided narrative and a narrative of your own creation.

You now have the foundation to go from examining a site record to creating tabulated data and sharing that data through a data narrative.

We've reached the end of this Digital Data Story. We hope that over the course of the two sections of this tutorial we've helped to build your understanding on the importance of data literacy, and have provided the chance to practice with going from forms to notes to tables to narratives.

Remember, the methods in this tutorial aren't limited to this data set. What we've discussed can be adapted and used on your own project data, or on any open data. (Like that at Open Context, hint, hint!) We'd love to see how these methods are used outside of this tutorial, and any data narratives you've created out in the world. Please get in touch and share!

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